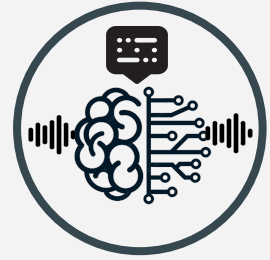


Audio Signal Processing for Decoding Morse Code with Neural Networks

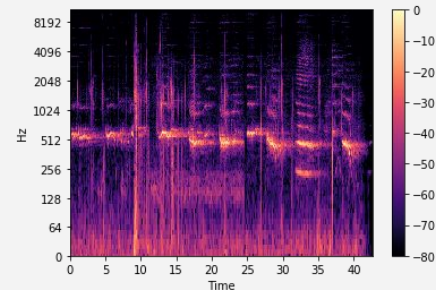


Introduction

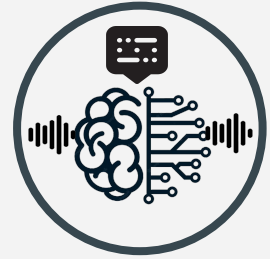
- Purpose: Decode Morse from audio into plain text
- Scope: Morse decoding utility as a command-line application
- Methodology:
 - Preprocessing audio data into smaller segments
 - Using Fast Fourier Transformation and Mel Spectrograms
 - Getting human-readable output



Converting Morse Code
from audio format
into plain text.



Morse Code

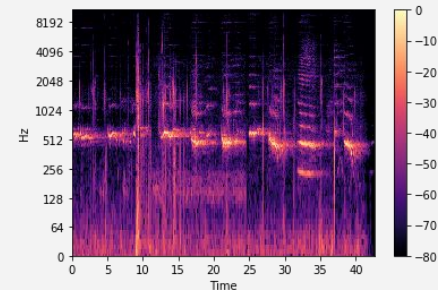


International Morse Code

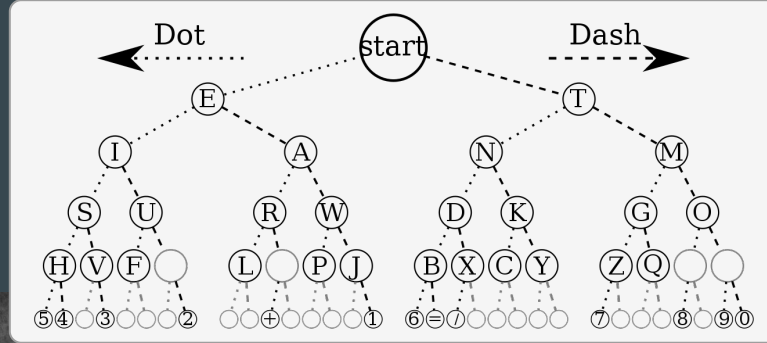
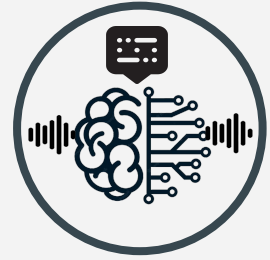
1. The length of a dot is one unit.
2. A dash is three units.
3. The space between parts of the same letter is one unit.
4. The space between letters is three units.
5. The space between words is seven units.

A	• —	U	• • —
B	— • • •	V	• • • —
C	— • — •	W	— • • —
D	— • •	X	— • • —
E	•	Y	— • — —
F	• • — •	Z	— — • •
G	— • •		
H	• • • •		
I	• •		
J	• — — —		
K	— • • —	1	• — — — —
L	• — • •	2	• • — — —
M	— —	3	• • • — —
N	— •	4	• • • • —
O	— — —	5	• • • • •
P	• — — •	6	— • • • •
Q	— • • —	7	— — • • •
R	• • • •	8	— — — • •
S	• • •	9	— — — — •
T	—	0	— — — — —

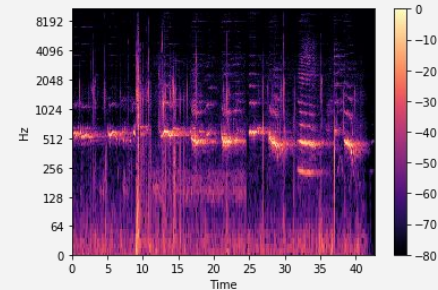
English characters
as dots and dashes.



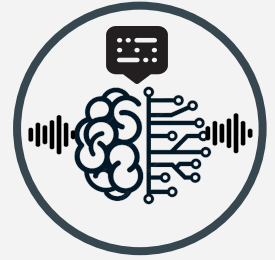
Traditional Methods



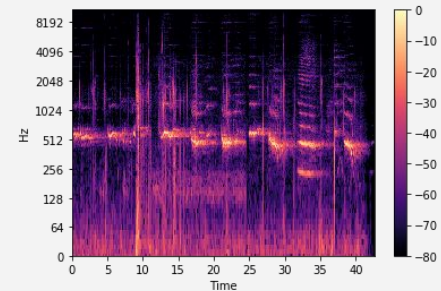
Decoded manually
and using binary trees.



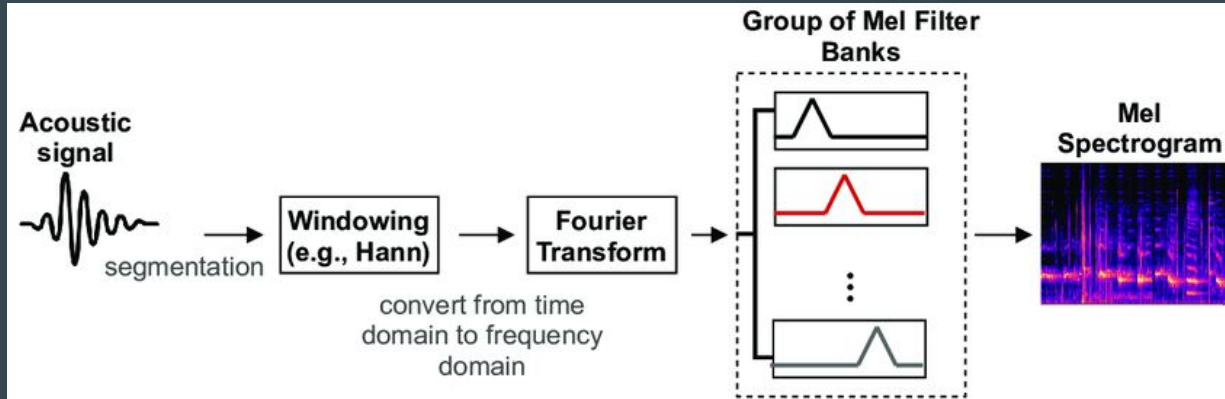
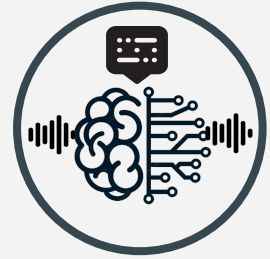
Why use a Neural Network?



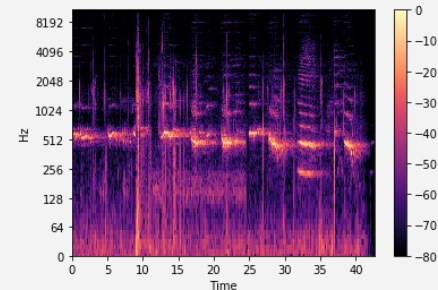
Pauses in between characters and words are not consistent.



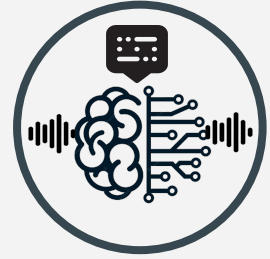
Mel Spectrograms



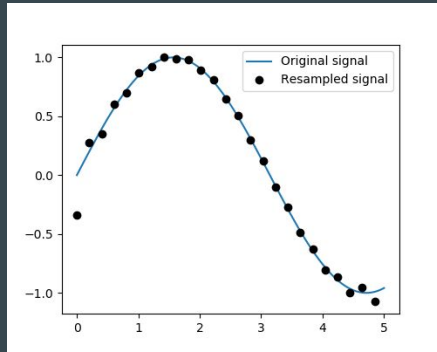
Using the peaks in the data to find dots and dashes.



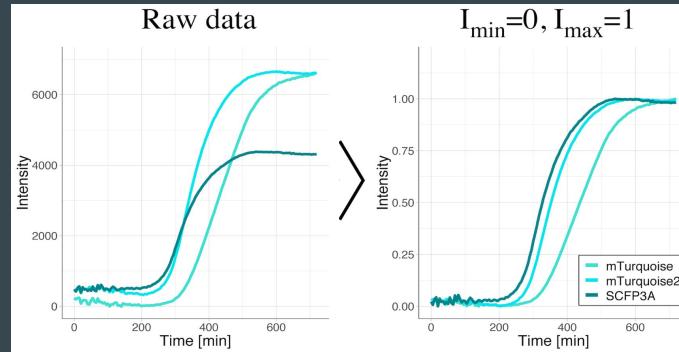
Preprocessing Data



- Resample - To increase accuracy
- Rescale - To facilitate prediction
- Generate Mel Spectrograms - For detection of dots and dashes.

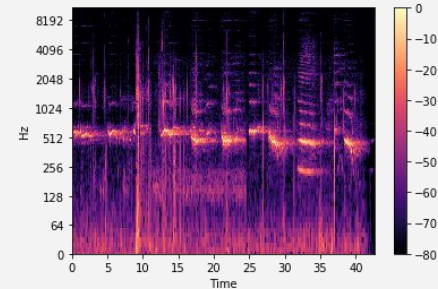


Resample

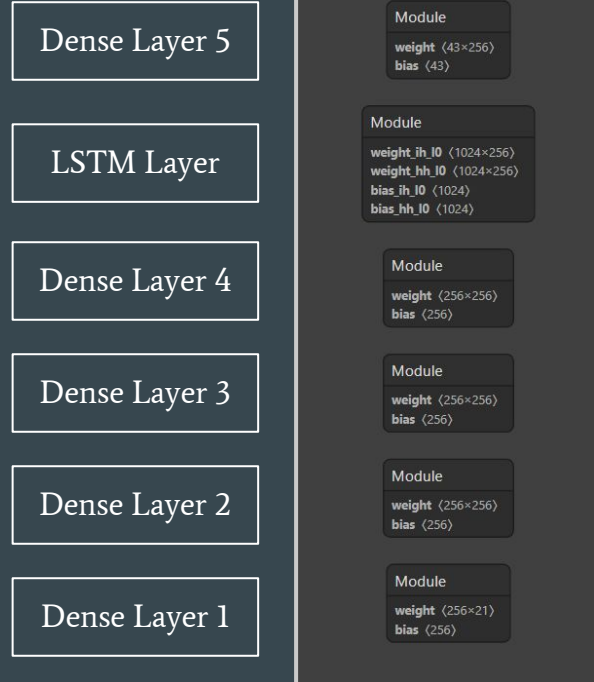
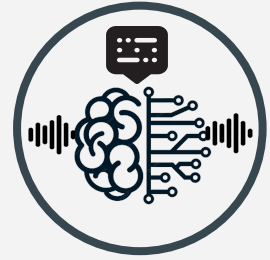


Rescale

Making data readable
for a neural network.

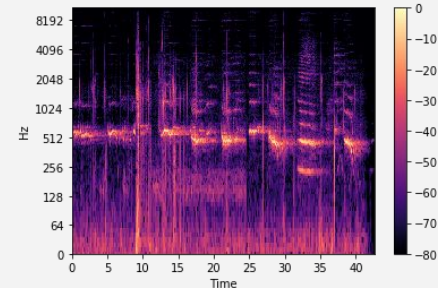


Neural Network Design

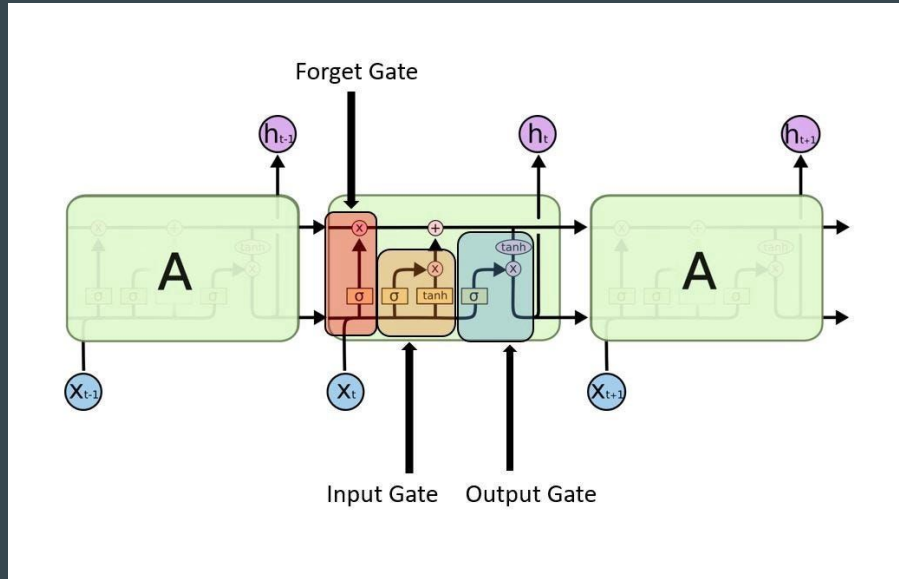
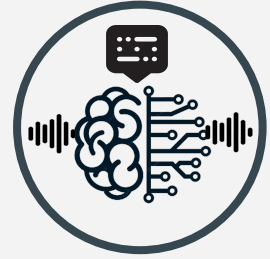


- Dense Layers are Linear Layers
- LSTM Layer is the most important here

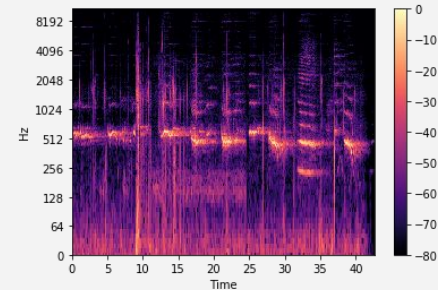
5 Linear Layers
1 LSTM Layer



LSTM



Decides on:
What to forget
What to Remember
What to output



Human-readable output

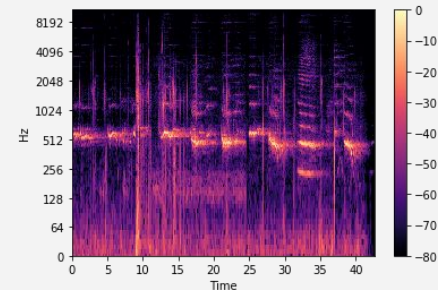
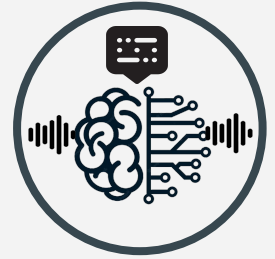
1 → { "1" : "A" } → A

Numerical output
from Neural Network
Range: [0...42]

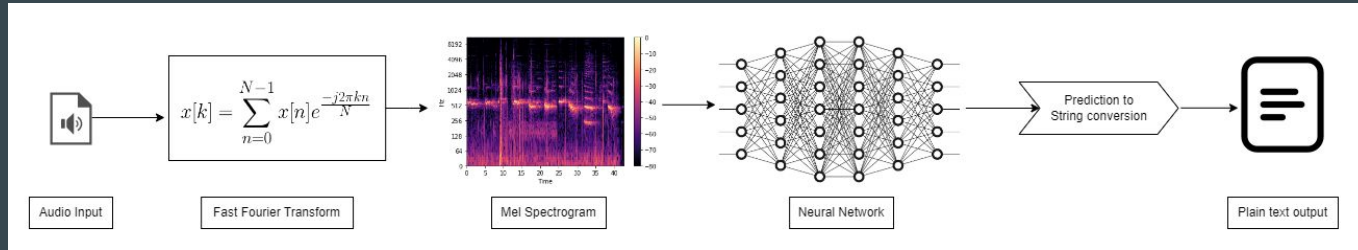
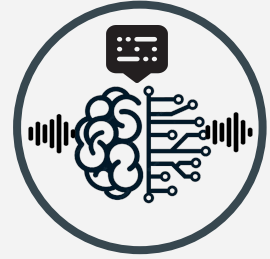
Pos Value
To Label

Printing Label values.

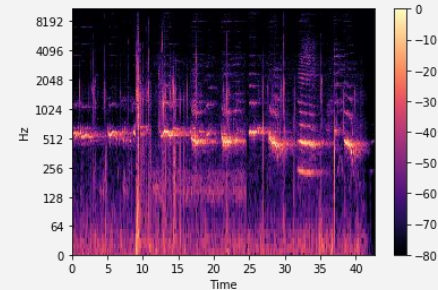
Converting numbers
into plain text.



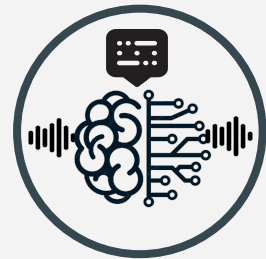
Overview



The whole picture.



Summary



- Audio signal processing in Machine Learning
- Understanding Morse Code
- Generating test data
- Converting audio to Mel Spectrograms
- Getting human-readable outputs

What have we explored?

